Week 7: In-Depth Reading – Chip Design

Week ILO: Interpret complex reading passages and demonstrate understanding of chip design terminology through glossary creation and contextual sentence use.

Session ILO: Identify and comprehend key stages of the chip design process and related technical terminology.

- **Learning Task 1:** Pre-reading Discussion & Prediction
- **Learning Task 2:** Guided Reading
- **Performance Task (Start):** Highlight & Record Key Vocabulary
- **Language Focus:**
  - Technical nouns (e.g., “parasitics,” “Systems-on-Chip”),
  - process verbs (“innovate,” “develop”).
  - and correct technical collocations.

First Hour: Guided Reading – Understanding Chip Design

- **Learning Task 1: Pre-reading Discussion & Prediction**
  View chip design process diagrams and predict possible stages/terms. → Small group discussion and whole-class sharing.

Useful Prediction Vocabulary

When previewing a diagram, you can think about:

Processes: *designing, simulating, verifying, optimizing, fabricating, etching, depositing, doping, testing, packaging, integrating*

Design Concepts / Operations: *modeling, routing, placing, connecting, layering, masking, checking, clustering, debugging*

Equipment / Tools: *wafer, mask, lithography system, deposition chamber, etcher, tester, probe station, packaging machine, cleanroom, EDA software*

Materials / Components: *silicon wafer, transistor, circuit, layout, photoresist, metal layer, dielectric, chip package*

Sequencing Connectors: first, next, then, after that, later, finally, subsequently, at this stage, in the final step

Example (Petroleum Refining Process)

If you see a diagram of crude oil entering a refinery, you might predict:

Specification and Planning – defining functions, performance goals, and design constraints

Logic and Circuit Design – using HDL (Hardware Description Language) to design circuits

Simulation and Verification – testing design behavior and fixing errors

Physical Design and Layout – placing components and checking design rules (DRC)

Fabrication – transferring the layout onto a silicon wafer using photolithography and etching

Packaging – protecting and connecting the chip for installation in devices

Testing and Validation – verifying functionality and yield before product release

Typical Stages in the Chip Design Process

Stage	What Happens	Key Technical Terms / Phrases
1 System requirement / Specification Definition	Specify demand / Define what the chip must do	Performance metrics System requeirements Power constraints Design constraints
2 Soc Architecture Planning	Decide which archecture is the best for the requirements	Bus architecture Micro architecture SoC
Integration Challenges 3 Logic Design	Design the logic gate, Design logic minimalzation, Convert function into logic circuit	HDL, VHDL RTL (Register transfer level) Logic synthesis
Design Innovation Required 4 Circuit Design	Planning the path of the wire, Find the simplest way to connect components	EDA Software Schematic Routing Corner analysis
5 Circuit Design / Simulation	Testing the design functionality using software, Test that design proforms as intended under different conditions.	DRC Analysis DFT Waveform analysis
Design verification 6 Physic Design / Layout	Vaildate the design is correct	VIAS
Chip Fabrication 7 Fabrication	To produce the chip	
8 Testing & Validation Packaging	To check the chip produced works fine	
9 Market release Test/Validation	Release the chip to the market	
10 Customer frrdback Integration and product release	Collect the suggestion from the customor	

- **Learning Task 2: Guided Reading**

Read an article describing key steps in chip design.

Instructions for Students

1. **Read** the passage carefully.
2. **Underline** or highlight *new or technical terms* (e.g., *System-on-Chip (SoC), dynamic range, parasitics, photolithography*).
3. **Annotate the margins** by summarizing each section or noting key ideas and reactions.
4. **Answer** the comprehension questions below.

Main Idea Questions

1. What is the main problem the semiconductor industry faces in chip verification?

   The huge numbers of error codes leading to an painful debug situation

2. How is artificial intelligence helping to solve this problem?

   AI can read lot of error code and summarize them to find what is the root cause

3. What is the overall message or conclusion of the article?

   AI can cut down the time of chip designing by many measures

Detail Questions

1. What does “EDA” stand for, and what is its role in chip design?

   Electronic Design Automation, it helps simulating the design before manufacture

2. What does DRC stand for, and why is it important?

   Design Rule Check, it check for vilation in circuit

3. How does Siemens’ Calibre Vision AI improve the verification process?

   It can summarize billions of error codes and summarize it to find the main problem

4. What benefits do AI-powered systems provide to less-experienced engineers?

   It can suggest the recommend debug path to the less-experiecned engineer for reference

5. What are two ways AI reduces the time needed for DRC analysis?

   Data rolling & Clustering

Sequencing Questions

1. Arrange these stages of change in the correct order:

   a. AI-powered clustering tools are introduced.

   b. Traditional DRC checking becomes a bottleneck.

   c. Chip designs become more complex.

   d. Siemens develops Calibre Vision AI.

   C B A D

2. Outline the general flow of how chip verification has evolved — from manual checking to AI-supported analysis.

- **Performance Task (Start): Highlight & Record Key Vocabulary**
  Select 10 technical terms/noun phrases from the text to build into a glossary in Session 2.
  Write definitions based on context and confirm with teacher or dictionary.

Second Hour: Glossary Creation and Application

Session ILO: Use chip design terminology accurately by producing a glossary with definitions and original example sentences.

- **Learning Task:** Review vocabulary logs and refine definitions
- **Performance Task:** Finalize and Submit Wafer Glossary
- **Language Focus:**
  - Word families (e.g., oxidize, oxidation),
  - Countable/uncountable usage (e.g., “a mask,” “some photoresist”),
  - Correct technical collocations.

- **Learning Task: Review vocabulary logs**

  Review vocabulary logs from Session 1 and refine definitions using Cambridge Learner’s Dictionary (or similar). Peer check with a partner to confirm meaning and sentence structure.

- **Performance Task: Finalize and Submit Chip Design Glossary**

  Complete a glossary with:

  - 10 chip design-related technical terms
  - A definition in your own words (confirmed for accuracy)
  - One original example sentence per word, demonstrating correct usage in an engineering context

Example:

Photolithography (n.) A process used in semiconductor manufacturing to transfer circuit patterns from a mask onto a silicon wafer using light and photoresist. Uncountable noun
Word family: Photolithographic (adj.) Photolithographically (adv.)
Technical collocations: Photolithography process Photolithography mask Deep ultraviolet (DUV) lithography Extreme ultraviolet (EUV) lithography Photolithographic patterning
Example Sentences: 1. <u>Photolithography</u> is one of the most critical steps in semiconductor fabrication because it defines the microscopic patterns of the circuit. 2. The <u>photolithographic</u> process defines the fine patterns that form transistors and interconnects on a chip. 3. The circuit patterns are <u>photolithographically</u> transferred onto the wafer surface through a patterned mask.

```
[ ] VIAS (穿洞)
[ ] DRC Analysis (Design rule check)
[ ] EDA Software
[ ] Register transfer level
[ ] HDL (Hardware Descriptive lang)
[ ] SoC (System on a chip)
[ ] Bus architecture
[ ] Logic synthesis
[O] Routing
[O] Design constraints
[ ] DFT (Design for Testibility)

Shift left
tape out
[-] Fabrication
[-] Optimize
[-] heat map
```